

# Final Boss Exam – Machine Learning (MDS + BDMA + MIRI)

Duration: 2h

*All questions have equal weight. Justify all your answers. Precision and clarity are essential.*

## Question 1

**True or False? Justify each answer (no marks for unjustified responses).**

- (a) Logistic regression always produces a linear decision boundary.
- (b) Lasso regression tends to select only one of many correlated features.
- (c) Decision trees require feature scaling to perform well.
- (d) Bagging increases model bias but decreases variance.
- (e) k-NN will always have 0 training error when  $k = 1$ .

## Question 2

Suppose you are given a dataset with 1000 points and 200 features. You apply logistic regression and observe high training accuracy but poor test accuracy.

- (a) What are the likely causes?
- (b) How would you adjust your model or training strategy to fix this?

## Question 3

You are given three models:

- (a) Logistic Regression
- (b) k-Nearest Neighbors
- (c) Multilayer Perceptron (MLP)

Rank them in terms of:

- Bias
- Variance
- Data efficiency

Provide justification based on theoretical and practical considerations.

## Question 4

In a binary classification task, the dataset is extremely imbalanced: 95% class A, 5% class B. You apply k-NN and get 95% accuracy.

- (a) Why is this misleading?
- (b) How would you modify the training or evaluation strategy to better reflect model performance?

## Question 5

Consider a 1-hidden-layer MLP with: - 2 input neurons - 1 hidden neuron (sigmoid activation) - 1 output neuron (sigmoid activation)

Suppose the input is  $\mathbf{x} = [1, 0]^T$ , weights from input to hidden are  $w_1 = [2, -3]$ , bias  $b_1 = 0.5$ , and hidden-to-output weight is  $w_2 = 1.0$ , bias  $b_2 = -1.0$ .

- (a) Compute the forward pass to obtain the final output  $\hat{y}$ .
- (b) Given true label  $y = 0$ , compute the derivative of the loss  $L = -y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})$  with respect to each weight using backpropagation.

## Question 6

Explain under which conditions logistic regression and a decision tree will give very different decision boundaries.

Provide:

- A sketched example
- A clear verbal description of why

## Question 7

You are comparing Gaussian Naive Bayes and logistic regression on the same dataset. The features are numeric and the conditional distributions are clearly non-Gaussian.

- (a) Which model do you expect to perform better and why?
- (b) How would you test your hypothesis empirically?

## Question 8

You are working with a model selection pipeline using 10-fold cross-validation. You find that model A has lower average validation error than model B, but a higher standard deviation across folds.

- (a) Which model would you choose and why?
- (b) Would your decision change if the dataset is small? Explain.

### Question 9

Consider regularized logistic regression. Define clearly the role of the regularization parameter  $\lambda$ .

Then, explain:

- How increasing  $\lambda$  affects decision boundary complexity
- Why cross-validation is necessary to select  $\lambda$
- What happens if  $\lambda$  is too small or too large

### Question 10

You are given a new dataset with the following properties: - High dimensionality ( $d = 500$ )  
- Small sample size ( $n = 100$ ) - Many correlated features - Unknown data distribution

- (a) Which models would you try first? Why?
- (b) How would you preprocess the features?
- (c) What strategies would you use to avoid overfitting?